

Numerical Analysis — Homework 4

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Deadline: May 22, 2025

1. Consider the function $f(x) = x^3$. The goal of this exercise is to solve the Gauss approximation problem

$$\min_{p \in \mathbb{P}_2} \int_0^1 |f(x) - p(x)|^2 e^{-x} dx ,$$

where $\mathbb{P}_2$ denotes the set of polynomials of degree ≤ 2 .

- (a) We define the weighted L^2 scalar product by

$$\forall \phi, \psi \in L^2([0, 1]), \quad \langle \phi, \psi \rangle = \int_0^1 \phi(x) \psi(x) e^{-x} dx.$$

Find an orthonormal basis of $\mathbb{P}_2$ with respect to this particular scalar product.

- (b) Solve the above Gauss approximation problem by using the above orthonormal basis.
 - (c) Plot your results for p and the function f on the interval $[0, 1]$.
2. Explain the main difference between Newton Cotes formulas and Gauss' quadrature.
 3. Explain how to apply Simpson's rule to find a numerical approximation of the integral

$$\int_{\Omega} e^{-\|x\|_2^2} dx$$

in $\mathbb{R}^3$ over the cylinder $\Omega = \{x \in \mathbb{R}^3 \mid x_1^2 + x_2^2 \leq 1, -1 \leq x_3 \leq 1\}$, where we use the notation $\|x\|_2 = \sqrt{x_1^2 + x_2^2 + x_3^2}$ to denote the Euclidean norm. Also derive an upper bound on the numerical approximation error.